

Research Interests

Machine learning, artificial intelligence for science, generative AI, numerical matrix and tensor algebra, statistical theory, quantum computing, approximation theory and algorithms, coding theory applications.

Positions

2018 - present	IBM Research , Yorktown Heights/Austin, USA <i>Senior Research Scientist</i> <i>Staff Research Scientist</i> <i>Research Staff Member</i> <i>Goldstine Postdoctoral Fellow</i>	June, 2025 - present May, 2022 - May, 2025 Jan, 2020 - May, 2022 Aug, 2018 - Dec, 2019
2022 - present	Oden Institute , University of Texas at Austin, TX, USA <i>Lecturer</i> <i>Visiting Researcher</i>	Jan, 2024 - present June, 2022 - present
2013 - 2018	University of Minnesota , Minneapolis, MN, USA <i>Research Assistant</i> , Department of Computer Science.	Sep, 2013 - May, 2018
2016 & 2017	Lawrence Berkeley National Laboratory , Berkeley, CA, USA <i>Research Intern</i> , Neural Systems and Data Science Lab <i>Research Intern</i> , Scientific Data Management group	June - Aug, 2017 May - Aug, 2016
2013	Seagate Technology , Shakopee, MN, USA <i>Signal Processing Intern</i>	May - Aug, 2013
2011 - 2012	Raman Research Institute , Bengaluru, India <i>Visiting Research Student</i>	June, 2011 - Aug, 2012

Education

2012-18	University of Minnesota , Minneapolis, MN, USA <i>Ph.D. Computer Science</i> Advisor: Yousef Saad Thesis: <i>Algorithmic advances in learning from large dimensional matrices and scientific data</i> <i>M.S. Computer Science</i> <i>M.S. Electrical Engineering</i> Advisors: Yousef Saad and Arya Mazumdar Thesis: <i>Randomized techniques for matrix decomposition and estimating the approximate rank of a matrix</i>	May 2018 October 2015 November 2014
2008-12	M.S. Ramaiah Institute of Technology , Bengaluru, India <i>B.Eng. Electronics and Communication</i>	

Publications

Journal articles

- [21] *Interpretable epistemic uncertainty attribution for decision-making under learned-model uncertainty*
R. Kaluarachchi, **S. Ubaru**, M. Zimon, D. Huh, R. Sawko, L. Horesh, and Y. Bengio,
Nature Machine Intelligence, accepted, 2026.
- [20] *Analysis of Power Iteration Algorithm with Partially Observed Matrix-vector Products*
S. Ghosh, L. Horesh, V. Kalantzis, Y. Lu, T. Nowicki, and **S. Ubaru**
Numerical Linear Algebra and Applications (NLAA), 33, no. 5: e70114, 2026.
- [19] *Counting Triangles of Graphs via Randomized Trace Estimation with Incomplete Matrix-Vector Products*
S. Ghosh, L. Horesh, V. Kalantzis, Y. Lu, T. Nowicki, and **S. Ubaru**
Monte Carlo Methods and Applications, vol. 32, no. 3, pp. 233-248, 2026.
- [18] *Advances in Analog Architectures and Randomized Algorithms: Computational and Energy Efficiencies*
V. Kalantzis, M. Squillante, **S. Ubaru**, T. Gokmen, C. Wu, A. Gupta, H. Avron, T. Nowicki, M. Rasch, M. Onen,
V. Lopez, E. Leobandung, Y. Kohda, W. Haensch, and L. Horesh
ACM Computing Surveys, 58 (14), 1-36, 2026.

- [17] *PCENet: High Dimensional Surrogate Modeling for Learning Uncertainty*
P.F. Shustin, **S. Ubaru**, M. Zimon, S. Lu, V. Kalantzis, L. Horesh, and H. Avron
SIAM/ASA Journal on Uncertainty Quantification (JUQ), 14 (1), 168-196, 2026
- [16] *Capacity Analysis of Vector Symbolic Architectures*
K.L. Clarkson, **S. Ubaru**, and E. Yang
Journal of Artificial Intelligence Research (JAIR), 86(32), 2026.
- [15] *Quantum and classical Monte Carlo algorithms for estimating Betti numbers of clique complexes*
I. Akhalwaya, A. Bhayat, A. Connolly, S. Herbert, L. Horesh, J. Sorci, and **S. Ubaru**
Quantum Journal, 9, 1901, 2025.
- [14] *Combinatorial Multi-armed Bandits: Arm Selection via Group Testing*
A. Mukherjee, **S. Ubaru**, K. Murugesan, K. Shanmugam, Ali Tajer
Transactions on Machine Learning Research (TMLR), 2835-8856, 2025
- [13] *Multivariate trace estimation using quantum state space linear algebra*
L. Yosef, **S. Ubaru**, L. Horesh, and H. Avron
SIAM Journal on Matrix Analysis and Applications (SIMAX), 46(1), 172–209, 2025.
- [12] *Randomized matrix-free quadrature: unified and uniform bounds for stochastic Lanczos quadrature and the kernel polynomial method*
T. Chen, T. Trogdon, and **S. Ubaru**
SIAM Journal on Scientific Computing (SISC), 47 (3), A1733-A1757, 2025
- [11] *Single-Pass Top-N Subgraph Centrality of Graphs via Subspace Projections*
V. Kalantzis, G. Kollias, **S. Ubaru**, L. Horesh and N. Abe
Journal of Complex Networks 13 (1), cnae049, 2025
- [10] *Representation of the Fermionic Boundary Operator*
I.Y. Akhalwaya, Y.H. He, L. Horesh, V. Jejjala, W. Kirby, K. Naidoo, and **S. Ubaru**
Physical Review A, volume 106, 022407, 2022.
- [9] *Dynamic graph and polynomial chaos based models for contact tracing data analysis and optimal testing*
S. Ubaru, G. Cohen, and L. Horesh
Journal of Biomedical Informatics, Novel Informatics Approaches to COVID-19 Research, 122, 103901, 2021.
- [8] *Spectrum-Adapted Polynomial Approximation for Matrix Functions*
L. Fan, D. Shuman, **S. Ubaru**, and Y. Saad
Algorithms, Special Issue: Efficient Graph Algorithms in Machine Learning, 13(11), 295, 2020.
- [7] *Sampling and multilevel coarsening algorithms for fast matrix approximations*
S. Ubaru and Y. Saad
Numerical Linear Algebra with Applications 26.3 : e2234, 2019.
- [6] *Fast estimation of $\text{tr}(f(A))$ via Stochastic Lanczos Quadrature*
S. Ubaru, J. Chen, and Y. Saad
SIAM Journal on Matrix Analysis and Applications (SIMAX), 38(4), 1075–1099, 2017.
- [5] *Low rank approximation and decomposition of large matrices using error correcting codes*
S. Ubaru, A. Mazumdar, and Y. Saad
IEEE Transactions on Information Theory, 63(9), 5544–5558, 2017.
- [4] *Formation enthalpies for transition metal alloys using machine learning*
S. Ubaru, A. Miedlar, Y. Saad, and J R. Chelikowsky
Physical Review B, (Vol.95, No.21), 2017.
- [3] *Fast estimation of approximate matrix ranks using spectral densities*
S. Ubaru, Y. Saad, and A.-K. Seghouane
Neural Computation, 29(5):1317–1351, 2017.
- [2] *Improving the Incoherence of a Learned Dictionary via Rank Shrinkage*
S. Ubaru, A.-K. Seghouane, and Y. Saad
Neural Computation, 29(1):263–285, 2017.
- [1] *Displaying gray scales by cross pairing select and data voltages in multi-line addressed LCD*
S. Ubaru and T.N. Ruckmongathan
IEEE Journal of Display Technology, 8(11), 669–677, 2012.

Conference proceedings

- [24] *Transformers Learn Faster with Semantic Focus*
P. Ram, K. Clarkson, T. Klinger, **S. Ubaru**, and A. Gray
Advances in Neural Information Processing Systems (NeurIPS), 2025.
- [23] *Iterative Solution of Sparse Linear Systems on Hybrid Architectures Using Homomorphic Encryption*
L. Horesh, V. Kalantzis, B. Trager, and **S. Ubaru**
IEEE High Performance Extreme Computing (HPEC), 2025.

- [22] *Power iteration with probabilistic updates for systems with heterogeneous performance*
S. Ghosh, L. Horesh, V. Kalantzis, G. Kollias, Y. Lu, T. Nowicki, and **S. Ubaru**
IEEE High Performance Extreme Computing (HPEC), 2025.
- [21] *On Efficient Solutions of General Structured Markov Processes in Quantum Computational Environments*
V. Kalantzis, M. S. Squillante, and **S. Ubaru**
IFIP Performance Conference, 2025.
- [20] *Topological Data Analysis on Noisy Quantum Computers*
I. Akhalwaya*, **S. Ubaru***, K. Clarkson, M. Squillante, V. Jejjala, Y. He, K. Naidoo, V. Kalantzis, and L. Horesh.
International Conference on Learning Representations (ICLR), 2024. (**Oral presentation, 1.2% of submissions**)
- [19] *Asynchronous Randomized Trace Estimation*
V. Kalantzis, **S. Ubaru**, G. Kollias, C.W. Wu, and L. Horesh.
International Conference on Artificial Intelligence and Statistics (AISTATS), 2024.
- [18] *Solving Sparse Linear Systems via Flexible GMRES with In-Memory Analog Preconditioning*
V. Kalantzis, M.S. Squillante, C. W. Wu, A. Gupta, **S. Ubaru**, T. Gokmen, and L. Horesh
IEEE High Performance Extreme Computing (HPEC), 2023.
- [17] *Quantum Graph Transformers*
G. Kollias, V. Kalantzis, T. Salonidis, and **S. Ubaru**
International Conference on Acoustics, Speech, and Signal Processing (ICASSP), 2023.
- [16] *Accelerating matrix trace estimation by Aitken’s Δ^2 process*
V. Kalantzis, G. Kollias, **S. Ubaru**, and T. Salonidis
International Conference on Acoustics, Speech, and Signal Processing (ICASSP), 2023.
- [15] *Efficient Scaling of Dynamic Graph Neural Networks*
V. Chakaravarthy, S. Pandian, S. Rajee, Y. Sabharwal, T. Suzumura, and **S. Ubaru**
Supercomputing (SC21), 2021.
- [14] *Analysis of stochastic Lanczos quadrature for spectrum approximation*
T. Chen, T. Trogdon, and **S. Ubaru**
International Conference on Machine Learning (ICML), 2021. (**Long presentation, 3% of submissions**)
- [13] *Projection techniques to update the truncated SVD of evolving matrices*
V. Kalantzis, G. Kollias, **S. Ubaru**, A. Nikolakopoulos, L. Horesh, and K.L. Clarkson
International Conference on Machine Learning (ICML), 2021.
- [12] *Sparse graph based sketching for fast numerical linear algebra*
D. Hu, **S. Ubaru**, A. Gittens, K. Clarkson, L. Horesh, and V. Kalantzis
International Conference on Acoustics, Speech, and Signal Processing (ICASSP), 2021.
- [11] *Dynamic Graph Convolutional Networks Using the Tensor M-Product*
O. Malik, **S. Ubaru**, L. Horesh, M. Kilmer, and H. Avron
SIAM International Conference on Data Mining (SDM), 2021.
- [10] *Multilabel Classification by Hierarchical Partitioning and Data-dependent Grouping*
S. Ubaru, S. Dash, O. Gunluk, and A. Mazumdar
Advances in Neural Information Processing Systems (NeurIPS), 2020.
- [9] *Spectrum-Adapted Polynomial Approximation for Matrix Functions*
L. Fan, D. Shuman, **S. Ubaru**, and Y. Saad
International Conference on Acoustics, Speech, and Signal Processing (ICASSP), 2019.
- [8] *Find the dimension that counts: Fast dimension estimation and Krylov PCA*
S. Ubaru, A.-K. Seghouane, and Y. Saad
SIAM International Conference on Data Mining (SDM), 2019.
- [7] *Spectral Sums Beyond Fast Matrix Multiplication: Algorithms and Hardness*
C. Musco, P. Netrapalli, A. Siford, **S. Ubaru**, and D. P. Woodruff
Innovations in Theoretical Computer Science (ITCS), 2018.
- [6] *UoI-NMF_{cluster}: A Robust Nonnegative Matrix Factorization Algorithm for Noisy Data*
S. Ubaru, K. Wu, and K. E. Bouchard
IEEE International Conference on Machine Learning and Applications (ICMLA), 2017. (**Best Paper Award**)
- [5] *Union of Intersections (UoI) for Interpretable Data Driven Discovery and Prediction*
K. E. Bouchard, A. F. Bujan, F. Roosta-Khorasani, **S. Ubaru**, Prabhat, A. M. Snijders, J.-H. Mao, E. F. Chang, M. W. Mahoney, and S. Bhattacharyya
Advances in Neural Information Processing Systems (NeurIPS), 2017.
- [4] *Multilabel Classification with Group Testing and Codes*
S. Ubaru and A. Mazumdar
International Conference on Machine Learning (ICML), 2017.
- [3] *Fast methods for estimating the Numerical rank of large matrices*
S. Ubaru and Y. Saad
International Conference on Machine Learning (ICML), 2016.

- [2] *Group testing schemes from low-weight codewords of BCH codes*
S. Ubaru, A. Mazumdar, and A. Barg
IEEE International Symposium on Information Theory (ISIT), 2016.
- [1] *Low rank approximation using error correcting coding matrices*
S. Ubaru, A. Mazumdar, and Y. Saad
International Conference on Machine Learning (ICML), 2015.

Book chapters

- [2] *Quantum Computing Algorithms For Decision Making Under Uncertainty*
L. Horesh, K. Clarkson, V. Kalantzis, M. Squillante, S. Ubaru, and A. Abboud
Air Force Research Lab Technical Report, AFRL-RI-RS-TR-2021-121, 2021.
- [1] *Applications of trace estimation techniques*
S. Ubaru and Y. Saad
High Performance Computing in Science and Engineering, LNCS book series, vol. 11087, ch. 2, pp 19–33, 2018.

Workshop papers and posters

- [10] *AVSAD: Automating Vector Symbolic Architecture Discovery with Iterative Evolution*
D. Scott, D. Zubarev, M. Esposito, A. Shinnar, A. Rahimi, K. L. Clarkson, L. Horesh, M. Hersche, and S. Ubaru
CLR Workshop on Logical Reasoning of Large Language Models, 2026.
- [9] *Quantifying policy uncertainty in generative flow networks with uncertain rewards*
R. Kaluarachchi, R. Sawko, S. Ubaru, D. Huh, M. Zimon, and L. Horesh
NeurIPS - ML×OR Workshop, 2025.
- [8] *Fast Linear Solvers via AI-Tuned Markov Chain Monte Carlo-based Matrix Inversion*
A. Lebedev, W. Lee, S. Ghosh, O. Yaman, V. Kalantzis, Y. Lu, T. Nowicki, S. Ubaru, L. Horesh, V. Alexandrov
Supercomputing (SC25): ScalAH Workshop, 2025.
- [7] *Counting Triangles of Graphs via Matrix Partitioning*
G. Kollias, V. Kalantzis, L. Horesh, S. Ubaru, and P. Traganitis
IEEE International Workshop on Machine Learning for Signal Processing (MLSP) 2024.
- [6] *On Quantum Algorithms for Efficient Solutions of General Classes of Structured Markov Processes*
V. Kalantzis, M. S. Squillante, and S. Ubaru
Workshop on Mathematical performance Modeling and Analysis (MAMA), 2024.
- [5] *Topological Data Analysis on Noisy Quantum Computers*
S. Ubaru, I Akhalwaya, K. Clarkson, and L. Horesh
Quantum Information Processing (QIP), 2023.
- [4] *On Quantum Algorithms for Random Walks in the Nonnegative Quarter Plane*
V. Kalantzis, M. S. Squillante, S. Ubaru, and L. Horesh
Workshop on Mathematical performance Modeling and Analysis (MAMA), 2022.
- [3] *Multilabel prediction in log time and data-dependent grouping*
S. Ubaru, S. Dash, O. Gunluk, and A. Mazumdar
NeurIPS - Workshop on Information Theory and Machine Learning, 2019.
- [2] *Tensor graph neural networks for prediction on time varying graphs*
O. Malik, S. Ubaru, L. Horesh, M. Kilmer, and H. Avron
NeurIPS - Workshop on Graph Representation Learning, 2019.
- [1] *Union of Intersections (UoI) for interpretable data driven discovery and prediction in neuroscience*
K. E. Bouchard, P. Sachdeva, S. Bhattacharyya, M. Balasubramanian, and S. Ubaru
Computational and Systems Neuroscience (Cosyne), 2019.

Preprints

- [5] *Closing the Consistency Gap: Self-Evolving Agents That Learn to Stay on Course*
E. Duesterwald, B. Elder, L. Ngweta, S. Ubaru, and M. Zimon, 2026.
- [4] *Exact Symmetry as Algebra: Group-Structured Tensors that Enforce Physical Selection Rules*
P. Hoyos, S. Ubaru, D. Huh, V. Kalantzis, K. L. Clarkson, M. Kilmer, H. Avron, and L. Horesh, 2026.
- [3] *Quantum Topological Data Analysis with Linear Depth and Exponential Speedup*
S. Ubaru, I. Akhalwaya, M. Squillante, K. Clarkson, and L. Horesh, 2021.
- [2] *Unsupervised Hierarchical Graph Representation Learning with Variational Bayes*
S. Ubaru and J. Chen, 2020.
- [1] *Provably convergent acceleration in factored gradient descent with applications in matrix sensing*
T Ajayi, D Mildebrath, A. Krylidis, S. Ubaru, G. Kollias, and K. E. Bouchard, 2019.

Patents

- [36] *P202502841 - Learning Discrete Group Symmetries from Data by Leveraging Structure of Convolutions*
P. Hoyos, **S. Ubaru**, D. Huh, V. Kalantzis, L. Horesh, and K. Clarkson. (filed)
- [35] *P202500687 - Incremental updating of truncated higher-order singular value decompositions*
D. Huh, L. Horesh, **S. Ubaru**, G. Kollias, and V. Kalantzis. (filed)
- [34] *P202502012 - Variational autoencoders for evaluating dynamical systems under uncertainty*
M. Zimon, **S. Ubaru**, L. Horesh, and V. Kalantzis. (filed)
- [33] *P202500571 - Straggler-tolerant determination of eigenvector in distributed computing architecture*
S. Ghosh, L. Horesh, V. Kalantzis, Y. Lu, **S. Ubaru**, and T. Nowicki. (filed)
- [32] *P202502080 - Power iteration with probabilistic updates for systems with heterogeneous performance*
V. Kalantzis, L. Horesh, T. Nowicki, S. Ghosh, **S. Ubaru**, G. Kollias, and Y. Lu. (filed)
- [31] *P202400952 - Transfer learning by network elasticity analysis*
L. Horesh, M. Zimon, K. Clarkson, M. Wegman, **S. Ubaru**, F. Martelli, and K. Yeo. (filed)
- [30] *P202502284 - Iterative solution of sparse linear systems on hybrid cloud*
V. Kalantzis, L. Horesh, **S. Ubaru**, and B. Trager. (filed)
- [29] *P202401590 - Machine learning from multi-dimensional data using tensor random forest models*
S. Ubaru, L. Horesh, K. Clarkson, K. Rhrissorakrai, A. Saenz, and L. Parida. (filed)
- [28] *P202400126 - Extracting nodes via alpha centrality with multiple damping factors*
V. Kalantzis, L. Horesh, **S. Ubaru**, and G. Kollias. (filed)
- [27] *P202203782 - Image compression using mixed-precision matrix sketching*
V. Kalantzis, T. Gokmen, **S. Ubaru**, and L. Horesh. (filed)
- [26] *P202202750 - Efficient techniques to identify the most influential nodes in graph structures*
L. Horesh, **S. Ubaru**, V. Kalantzis, and G. Kollias. (filed)
- [25] *US20250258826A1 - Efficient look-up for vector symbolic architectures (VSA)*
T. Gokmen, V. Kalantzis, P. Ram, C. Wu, K. Clarkson, L. Horesh, and **S. Ubaru**. (pending)
- [24] *US20250181465A1 - Asynchronous modeled system for fault-tolerant environments*
L. Horesh, V. Kalantzis, G. Kollias, **S. Ubaru**, and C. Wu. (pending)
- [23] *US20250190602A1 - Privacy-preserving graph analytics on hybrid cloud environments*
V. Kalantzis, L. Horesh, **S. Ubaru**, and G. Kollias. (pending)
- [22] *US20250036987A1 - Quantum Graph Transformers*
G. Kollias, V. Kalantzis, T. Salonidis, and **S. Ubaru**. (pending)
- [21] *US20240370524A1 - Aitken acceleration for estimating electronic structures of materials*
V. Kalantzis, G. Kollias, **S. Ubaru**, and T. Salonidis. (pending)
- [20] *US20240320033A1 - Processing sparse linear systems using distributed resources*
V. Kalantzis, **S. Ubaru**, and L. Horesh. (pending)
- [19] *US20240176843A1 - Solving systems of linear equations using mixed precision*
T. Gokmen, V. Kalantzis, **S. Ubaru**, and L. Horesh. (pending)
- [18] *US20240135185A1 - High dimensional surrogate modeling for learning uncertainty*
P. Fink, **S. Ubaru**, H. Avron, V. Kalantzis, and L. Horesh. (pending)
- [17] *US20240020565A1 - Quantum circuit for estimating matrix spectral sums*
S. Ubaru, I. Akhalwaya, K. Clarkson, M. Squillante, V. Kalantzis, and L. Horesh. (pending)
- [16] *US20240022247A1 - Quantum circuit for pairwise testing*
S. Ubaru, I. Akhalwaya, K. Clarkson, M. Squillante, V. Kalantzis, and L. Horesh. (pending)
- [15] *US20240037304A1 - Quantum circuit for simulating boundary operator*
I. Akhalwaya, Y. He, L. Horesh, V. Jejalla, W. Kirky, K. Naidoo, and **S. Ubaru** (pending)
- [14] *US20230410113A1 - Detecting network patterns using random walks*
K.P. Onak, **S. Ubaru**, A. Abboud, and T. Suzumura. (pending)
- [13] *US20220300575A1 - Determining triangles in graph data structures using crosspoint array*
V. Kalantzis, **S. Ubaru**, L. Horesh, and H. Avron. (pending)
- [12] *US12579459B2 - A short-depth quantum system for entanglement with count register*
I. Akhalwaya, **S. Ubaru**, K. Clarkson, M. Squillante, V. Kalantzis, and L. Horesh.
- [11] *US12561594B2 - Log-depth quantum circuits for matrix trace estimation*
S. Ubaru, K. Clarkson, I. Akhalwaya, M. Squillante, V. Kalantzis, and L. Horesh.
- [10] *US12505370B2 - Linear-depth quantum system for Topological Data Analysis*
I. Akhalwaya, **S. Ubaru**, K. Clarkson, M. Squillante, V. Kalantzis, and L. Horesh.

- [9] *US12499100B2 - System to update the most influential nodes of dynamic graphs*
V. Kalantzis, G. Kollias, **S. Ubaru**, L. Horesh, and N. Abe.
- [8] *US12118059B2 - Projection-based techniques for updating SVD in evolving datasets*
V. Kalantzis, G. Kollias, **S. Ubaru**, L. Horesh, and K. Clarkson.
- [7] *US11907715B2 - Hardware acceleration with preconditioners*
V. Kalantzis, L. Horesh, and **S. Ubaru**.
- [6] *US11790033B2- Accelerated quasi-newton methods on analog crossbar hardware*
V. Kalantzis, **S. Ubaru**, L. Horesh, H. Avron, and M. Onen.
- [5] *US11657312B2 - Short-depth quantum amplitude estimation without eigenstate collapse*
I. Akhalwaya, K. Clarkson, L. Horesh, M. Squillante, **S. Ubaru**, and V. Kalantzis.
- [4] *US11520855B2 - Matrix sketching using analog crossbar architectures*
L. Horesh, M. Onen, H. Avron, T. Gokmen, **S. Ubaru**, and V. Kalantzis.
- [3] *US11500963B2 - Fast PCA of evolving data using analog crossbar array*
S. Ubaru, V. Kalantzis, L. Horesh, M. Squillante, and H. Avron.
- [2] *US11386507B2 - Tensor-based predictions from analysis of time-varying graphs*
O. Malik, **S. Ubaru**, L. Horesh, M. Kilmer, and H. Avron.
- [1] *US11379758B2 - Automatic multilabel classification using machine learning*
S. Ubaru, S. Dash, O. Gunluk, A. Mazumdar and L. Horesh.

Presentations

- 2026
 - *Capacity Analysis of Vector Symbolic Architectures*
 - Randomized Numerical Linear Algebra Workshop, ICERM, Brown University, RI
 - *Transformers Learn Faster with Semantic Focus*
 - Capital One Panel Presentation
- 2025
 - *Capacity Analysis of Vector Symbolic Architectures*
 - Computational Biosciences Seminar, Lawrence Berkeley National Laboratory, CA
 - *Career Pathways* Panelist
 - Rising Stars workshop, UT Austin
- 2024
 - *Quantum Graph Transformers*
 - SIAM Conference on Applied Linear Algebra (LA24), Paris France
 - *Topological data analysis on noisy quantum computers*
 - SIAM Conference on Applied Linear Algebra (LA24), Paris France
 - International Conference on Learning Representations (ICLR), Vienna, Austria
 - *Asynchronous Randomized Trace Estimation*
 - International Conference on Artificial Intelligence and Statistics (AISTATS), Valencia, Spain
 - *Career Pathways* Panelist
 - Rising Stars workshop, UT Austin
- 2023
 - *Dynamic graph representation learning using tensor algebra*
 - UTD CS Colloquium, University of Texas Dallas
 - *Topological data analysis on noisy quantum computers*
 - QuSoft, CWI, Amsterdam, The Netherlands
 - QIP 2023 conference, Ghent, Belgium
 - *Algorithms for estimating spectral sums for large matrices*
 - Dr. Rachel Ward's group, University of Texas at Austin
- 2022
 - *Industry Panelist*
 - SIAM Conference on Mathematics of Data Science (MDS22), San Diego
 - *Towards quantum advantage on noisy computers*
 - Google Research, Bengaluru
 - *Dynamic graph representation learning using tensor algebra*
 - Oden Institute Seminar Series, University of Texas at Austin
 - SIAM Conference on Mathematics of Data Science (MDS22), San Diego
 - Data Science group, University of California, San Diego.
 - *Multilabel Classification by Hierarchical Partitioning and Grouping*
 - RISE workshop, Austin, TX
 - *Quantum Topological Data Analysis with Linear Depth and Exponential Speedup*
 - Quantum Information Group, University of Texas at Austin
 - Quantum computing course, Columbia University, New York

- 2021 • *Into another dimension - a new tensor algebra for learning on dynamic graphs*
– Seminar at University of Albany, NY
- *Algorithms for estimating spectral sums for large matrices*
– Oden Institute Seminar Series, University of Texas at Austin
- *Novel Tensor Framework for Data Representation and Compression*
– SIAM Conference on Computational Science and Engineering (CSE21)
- 2020 • *Multilabel Classification by Hierarchical Partitioning and Data-dependent Grouping*
– Neural Information Processing Systems (NeurIPS) Virtually held
- *Novel tensor framework for model reduction and neural networks*
– ICERM: Workshop on Algorithms for Dimension and Complexity Reduction, Brown University, RI
- 2019 • *Tensor graph neural networks for prediction on time varying graphs*
– NeurIPS: Workshop on Graph Representation Learning, Vancouver, Canada
- *Multilabel prediction in log time and data-dependent grouping*
– NeurIPS: Workshop on Information Theory and Machine Learning, Vancouver, Canada
- *Spectrum approximation by Lanczos Quadrature and Preconditioned SVRG*
– 20th International Conference On Preconditioning, Minneapolis, MN, USA
- *Find the dimension that counts: Fast dimension estimation and Krylov PCA*
– SIAM International Conference on Data Mining (SDM), Calgary, Canada
- *Error Correcting Codes for Machine Learning.*
– CSA Seminar, Indian Institute of Science (IISc), Bengaluru, India
- 2018 • *Error Correcting Codes for Machine Learning*
– Theory Seminar, University of Massachusetts, Amherst, MA
- IP Seminar, IBM T.J. Watson Research Center, NY
- 2017 • *UoI-NMF_{cluster}: Robust Nonnegative Matrix Factorization Algorithm for Noisy Data*
– International Conference on Machine Learning and Applications (ICMLA), Cancun, Mexico
- *Multilabel Classification with Group Testing and Codes*
– Neural Systems and Engineering Labs, Lawrence Berkeley National Laboratory, CA
- International Conference on Machine Learning (ICML), Sydney, Australia
- *Error Correcting Codes for Machine Learning*
– The University of Melbourne, Melbourne, Australia
- *UoI-NMF_{cluster} and UoI-CUR: Union of Intersections methods for matrix approximations*
– Neural Systems and Engineering Labs, Lawrence Berkeley National Laboratory, CA
- 2016 • *Error correcting codes for low rank approximation and group testing*
– BLISS Seminar, University of California, Berkeley, CA
- *Fast methods for estimating the Numerical rank of large matrices*
– International Conference on Machine Learning (ICML), New York, NY
- 2015 • *Low rank approximation using error correcting coding matrices*
– International Conference on Machine Learning (ICML), Lille, France

Awards

- 2023 **Master Inventor Award**, IBM Research (3-year term)
- 2023 **Outstanding Technical Achievement Award**, IBM Research
- 2022 **IBM Research Accomplishment Award**
- 2022 **Outstanding Innovation Award**, IBM Research
- 2021 **IBM Research Accomplishment Award**
- 2019 **UMN Best Dissertation Award**, Computer Science Department nominee
- 2018 **Herman Goldstine Fellowship**, IBM Research
- 2017 **Best Paper Award**, International Conference on Machine Learning and Applications (ICMLA)
- 2011 **Visiting Research Student Program**, Raman Research Institute

Teaching

University of Texas, Austin, Instructor

- Spring 2025 CSE 392/CS 395T/M 397C: Matrix and Tensor Algorithms for Data
- Spring 2024 CSE 392: Matrix and Tensor Algorithms for Data

IBM Research, Co-instructor

- Summer 2023 Mathematics of Big Data: Sketching and (Multi-) Linear Algebra, Organized at IBM, Almaden
- Summer 2021 Mathematics of Big Data: Sketching and (Multi-) Linear Algebra at MSRI Berkeley

University of Minnesota, *Teaching Assistant*

Spring 2018 CSci 2033, Elementary Computational Linear Algebra, *Recitation Instructor*

Fall 2017 CSci 5304, Computational Aspects of Matrix Theory, *Teaching Assistant*

Spring 2017 CSci 8314, Sparse Matrix Computations, *Temporary Instructor*

Mentoring

IBM Research - Interns

2025 Ramón Nartallo-Kaluarachchi (PhD Student, Oxford University)

2024, 25 Paulina Hoyos (PhD Student, UT Austin)

2023 Liron Mor Yosef (PhD Student, Tel Aviv University)

2023 Arpan Mukerjee (PhD Student, RPI). Now: Postdoc, Imperial College London

2022 Elizabeth Yang (PhD Student, UC, Berkeley). Now: OpenAI

2021 Paz F. Sustin (PhD Student, Tel Aviv University). Now: Postdoc, Oxford University

2021 Lucky Yerimah (PhD Student, RPI). Now: Los Alamos National Lab

2021 Emily Thompson (REU Student via. DIMACS). Now: PhD student, SMU

2020 Dong Hu (PhD Student, RPI). Now: Walmart Labs

2019 Osman Malik (PhD Student, U of Colorado, Boulder). Now: Encube Technologies

PhD Thesis committee

2024 Dong Hu (PhD Student, RPI). Now: Walmart Labs

Services

Organizer: SIAM MDS 2024 - Member of organizing committee.

SIAM LA 2024 - Minisymposium on Quantum Numerical Linear Algebra

SIAM CSE 2023 - Minisymposium on Randomized Algorithms for Matrix and Tensor Analysis

IEEE ICMLA Challenge 2018 - Parts based decomposition of noisy data.

Reviewer: National Science Foundation (NSF) grant proposals, 2023.

Conferences: Neural Information Processing Systems (NeurIPS) 2016, 2019-25; International Conference on Machine Learning (ICML) 2018- 25; International Conference on Learning Representations (ICLR) 2021-26; Artificial Intelligence and Statistics (AISTATS) 2024-26; AAAI conference on Artificial Intelligence 2020-22 ; Conference on Uncertainty in Artificial Intelligence (UAI) 2019-25; ACM-SIAM Symposium on Discrete Algorithms (SODA) 2021, 25; Quantum Information Processing Conference (QIP) 2026; IEEE International Conference On Machine Learning And Applications 2018-19; IEEE International Symposium on Information Theory (ISIT) 2017.

Journals: SIAM Journal on Scientific Computing (SISC); SIAM Journal on Matrix Analysis and Applications (SIMAX); Acta Materialia; PLOS One; Journal of Machine Learning Research; IEEE Transactions on Signal Processing; IEEE Transactions on Image Porcessing; Signal Processing Letters, IEEE Transactions on Information Theory; Electronic Transactions on Numerical Analysis ; IEEE Transactions on Signal and Information Processing over Networks; BIT Numerical Mathematics; Linear Algebra and Its Applications; Journal of Mathematics and Applications, Mathematical Modeling and Analysis; Journal of Imaging.

Programming skills

Python, Pytorch, Matlab, C, C++, LATEX, Basic web programming.